Hate Speech Detection Using Brazilian Imageboards

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ABSTRACT

With the changes in human interaction prompted by the development of communications platforms over the internet, hate speech and offensive language emerged as a contemporary problem. Social networks allow users with different opinions and backgrounds to interact without direct eye-to-eye contact. It brings a sense of safety to promote hate speech, which is even more significant in anonymous environments. There are sites called imageboards, composed of different boards aggregating different topics. On some boards, anonymous users widely promote hate speech. However, only a few works in literature have focused on hate speech in imageboards content. This work aims to classify Brazilian Portuguese texts to detect hate speech, using data from the Brazilian 55chan imageboard to build a dataset with hate speech content. Three classifiers were trained to hate speech binary classification. The Linear Support Vector Classifier achieved the best result with 0.955 of F1-score.

CCS CONCEPTS

• **Human-centered computing → Social media**; • **Computing methodologies → Natural language processing**; **Information extraction**.

KEYWORDS

hate speech detection, imageboards, text mining

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1 INTRODUCTION

'Hate speech' (HS) is a growing contemporary problem, spreading in the traditional media, online and in political discourse, including sometimes in statements made by politicians and public officials [3, 23]. Given this issue, social networking platforms and other internet services have been making efforts to identify and censor problematic posts and regulate the interaction between people with different psychological and cultural backgrounds [26]. However, some users would rather interact anonymously using minimal text on image-posting forums known as imageboards (IB), avoiding such regulations.

IBs provides the ideal scenario for HS, manifested as detractive messages towards people or groups because of their race, ethnicity, gender, sexual orientation, nationality, religion, or other attributes [26]. Each IB site is structured in boards, aggregating different topics or subjects. For example, one of the most famous IBs, 4chan[1], has the well-known */pol/* (Politically Incorrect) board with controversial content, where users posting anonymously are not easily linked to their real-world identities and are more likely to use offensive language [25].

An eminent Brazilian IB is 55chan[2] [9]. Like 4chan, it also has a controversial */pol/* board, where users addresses global political issues in ways that strongly deviate from what is considered 'politically correct' understanding and from cultural inclusion [13]. Within the */pol/* board lies a misogyny filled board called */escoria/*, abounding HS against women.

Studies recently paid attention to 4chan's */pol/* board, in regard to HS detection[13, 17, 28]. Some researchers focused on the alt-right movement and its effects on web [11, 27]. Other studies focused on users' mental health [16]. However, there are few studies using IB data from 55chan and other boards in Brazilian Portuguese language, along with a lack the development of methods for analyzing text in languages other than English [21].

This work aims to classify Brazilian Portuguese texts to detect HS. To accomplish this, it has as secondary objectives: (1) build an HS dataset from offensive 55chan boards; (2) collect tweets to fill the HS dataset; (3) create a baseline for offensive language classification using this HS dataset.

The experiment consists in treating HS detection as a binary classification problem. Thus, three classifiers were trained. The results were promising, achieving 0.955 of F1-score in the LSVC classifier.

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[1] 4chan.org
[2] http://www.55chan.org

The remainder of this work is structured as follows. Section 2 describes some related work that addresses offensive and HS detection. Section 3 describes the LIWC dictionary, used in for data filtering and data visualization. Section 4 describes the classifiers used in this work to HS detection. Section 5 explains the Data Collection and Filtering, Preprocessing, Data Visualization, Feature Preparation, and Classifiers Training tasks executed in this work. Section 6 provides the results. Finally, Section 7 summarizes our contributions in this work.

2 RELATED WORK

This section describes some related work that addresses offensive and HS detection. For the best of our knowledge, there are few works in literature in subjective analysis and sentiment analysis that focuses on tasks related to HS and offensive language. This limitation is even greater for IBs and anonymous data on Brazilian Portuguese language.

Almeida et al. [3] proposed an approach for HS detection based on Information Theory quantifiers (entropy and divergence) to represent documents. The proposed approach, captured weighted information of words, rather than just their frequency in documents. They used three classification algorithms: Multinomial Naive Bayes, K-Nearest Neighbor and Maximum Entropy. The documents for classification were categorized in hateful, offensive and regular classes. For the TF-IDF representation, the classifier that obtained the best effectiveness was Maximum Entropy.

Watanabe et al. [26] proposed a method to classify tweets with offensive and hate content. They collected and combined three datasets with tweets into a larger dataset. Each tweet was assigned to one of three classes: *clean*, *offensive* and *hateful*. Next, they proposed a method to extract four types of features: sentiment-based, semantic, unigram and pattern features. Then, they trained classifiers for binary and ternary classification problems using three classification algorithms: Random Forest, SVM and J48graft. Their proposed approach reached an accuracy equal to 87.4% for the binary classification of tweets into offensive and non-offensive, and an accuracy equal to 78.4% for the ternary classification of tweets into, hateful, offensive and clean.

Rodriguez et al. [22] proposed a semi-automatic way to detect sensitive topics that generate HS on Facebook and a framework to cluster posts to detect HS. Then, K-means clustering was applied to determine the most discussed topics. The clusters and parameters were empirically evaluated and, according to the authors, the results were promising.

This work differs from all aforementioned because it focuses on IBs, that involves 'chanspeak' (the language used in IBs) and hostile and anonymous environments. Moreover, this work focuses on the Brazilian Portuguese language.

3 LIWC DICTIONARY

LIWC is an application that counts words based on dictionary files. It was developed to study components from textual files and associate them with the emotional, cognitive, and structural aspects [19]. After analyzing texts, the program outputs a vectorial representation in which each attribute stores the percentage of words from a category or subcategory of the dictionary.

The standard LIWC dictionary uses a structure of categories and sub-categories. The main categories are associated to Linguistic Dimensions, like Function words, Common verbs, Numbers, Affective processes, Social processes, Cognitive processes, Perceptual processes, Biological processes, Drives, Time orientations, Relativity, Personal concerns, and Informal language [19]. LIWC has many versions, and the most recent was released in 2015. LIWC dictionary file has been adapted into many languages, including Brazilian Portuguese[3] [6].

4 CLASSIFIERS

Three classifiers were chosen based on the HS detection works of Alfina et al. [2] and Watanabe et al. [26]. The classifiers are Multinomial Naive Bayes (MNB), Random Forest (RF) [5] and Support Vector Machine (SVM) [7].

The Naive Bayes classifier is based on the Bayes theorem, which assumes the independence of random variables, and for document representation using discrete distributions of real numbers as attributes, the Multinomial Naive Bayes (MNB) is usually used [14]. The SVM is a supervised machine learning algorithm that aims to solve a classification problem by finding an optimal hyperplane in a high-dimensional vector space [7]. RF is also a supervised learning algorithm, however, it is based on a set of decision trees that receive random and independent population samples of the same distribution. Then, it performs a classification task where each prediction values one vote [5].

5 METHODOLOGY

The methodology proposed in this work follows six phases: (i) data collection from two data sources; (ii) data filtering using a lexicon; (iii) data preprocessing; (iv) data visualization; (v) feature preparation; and (vi) training of offensive language classifiers. The dataset built in this section is available on Github[4].

5.1 Data Collection

The data collection is executed from two particular data sources. The first data source is 55chan, and the second data source is Twitter. Data were then combined into a single dataset composed by textual content from the two sources since it's rare to find non-offensive posts in 55chan.

To represent the offensive language labeled documents, data collection from 55chan are from two selected boards, both crammed with slang, swear words, and disrespectful offensive words. One is the */pol/* (Politically Incorrect) board, which contains controversial political content, similar to the */pol/* board from 4chan. The second board is the */escoria/*, which, despite the purpose of being a space where users ask questions about love relationships, contains many offensive comments about women.

The 55chan, like many other chans, is made over the Tinyboard engine, currently forked to vichan[5]. This engine offers an API that allows, if enabled, to read all the catalog, boards and threads in JavaScript Object Notation (JSON) [8] format. Since 55chan enables

[3]LIWC lexicons are available for download in www.liwc.net/dictionaries.
[4]https://github.com/LaCAfe/Dataset-Hatespeech
[5]https://github.com/vichan-devel/vichan

this API, it was possible to collect data using it. Thus, all the catalog from both boards were obtained. Each catalog contains all the threads from one specific board. We then proceeded to collect all the posts from each thread.

The neutral labeled documents consisted of Twitter data, collected using the Twitter API[6]. Retweets (reposted tweets from another user) were unconsidered. The tweets were collected using no filter other than being in Portuguese. The samples were collected using the sample[7] method in API and the tweepy[8] Python library.

5.2 Data Filtering

The LIWC 2015 Brazilian Portuguese lexicon is used to filter the collected data. Similar approach is followed by Rodriguez et al. [22] using VADER [12] and JAMMIN [4] instead. The filters were based on LIWC categories for anger, anxiety, and positive emotions.

There are two different filters, one for each label: offensive and non-offensive. Tweets containing words from categories anxiety or anger were removed to assure that the data collected from Twitter were non-offensive. Similarly, to ensure that the data collected from 55chan boards contains offensive texts, posts with words from the anxiety and anger categories were considered and, consequently, posts with positive words were disregarded.

At the end of the data filtering step, the dataset contains 3836 offensive documents from 55chan. As the number of filtered neutral tweets are far superior, a simple random sampling was adopted to reduce the amount of non-offensive documents to 3836 tweets. This sampling is equal to the number of 55chan posts to build a balanced dataset.

5.3 Preprocessing

The preprocessing phase of the collected data aims to improve the quality of these data. Then, this data can be treated by classifiers. All the text was standardized to lowercase letters; all hyperlink was removed, and the sentences were tokenized.

The *tokenization* process is one of the first steps that divide the text into smaller units. These units, called tokens, can be words, sentences, phrases, and paragraphs [24]. This work uses tokens of words.

5.4 Data Visualization

The Data Visualization phase addresses data analysis from 55chan material, using the tokens extracted in the previous phase. To achieve it, word clouds were extracted from the */pol/* and */escoria/* boards. Word clouds are a straightforward and visually appealing visualization method for text [10]. They are used in various contexts as a means to provide an overview by distilling text down to those words that appear with highest frequency [10].

Some tokens were omitted to plot the word cloud, as they represent stopwords. They were filtered according to the *function word* and *verbs* categories from the 2015 version of Brazilian Portuguese LIWC lexicon. Nevertheless, these tokens were not discarded or hidden to the feature preparation or the classification training.

[6]https://developer.twitter.com/en/docs.html
[7]https://developer.twitter.com/en/docs/tweets/sample-realtime/overview/GET_statuse_sample.html
[8]https://www.tweepy.org/

Figure 1 shows the word cloud from 55chan. The words: *vida* (life), *anos* (years) are frequent because the board /escoria/ focus is to discuss about loving relationships and users tend to tell about their particular lives. The frequency of the word 'years' came from contexts that users are discussing about age, period or from sentences like: '*Temos como nos recuperar de anos de doutrinação marxista?*' (Can we recover from **years** of Marxist indoctrination?). Other words are also frequent: *depósito* (deposit), *po**a* (swear-word). The terms 'depósito' and 'po**a' occurs frequently together, forming the term 'depósito de po**a', a misogynist brazilian 'chanspeak' to refer women. There is a high frequency of the word 'judeus' (jews) in 55chan because there is a large inherent anti-semitism. Other words less frequent, like: *homem* (man), *estado* (state), *Brasil* (Brazil) and *país* (country) reveals the political nature of the board /pol/.

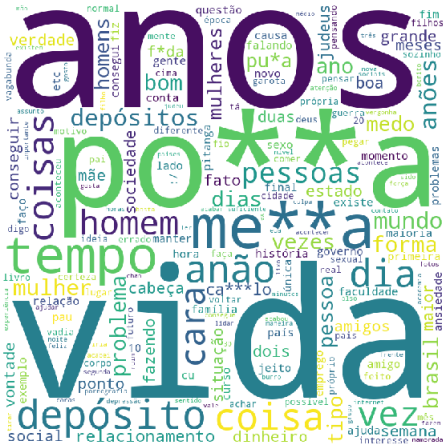

Figure 1: Word cloud representing the most frequent words in the /escoria/ and /pol/ 55chan boards.

5.5 Feature Preparation

In the feature preparation phase, the weighting scheme TF-IDF adopted. The main objective of discovering important words in an unstructured or semi-structured text [20]. A value is assigned to each term, indicating its importance. This value is calculated combining the term frequency and inverse document frequency.

The intuition of TF-IDF consists of filtering all the text to classify the words with values representing a degree of affinity with the text [20]. Terms that are common in a single or small group of documents tend to have a high TF-IDF concerning words that occur several times, such as articles, pronouns, and prepositions.

5.6 Classifiers Training

The three classifiers were trained and evaluated using the 10-fold cross-validation technique. The MNB classifier was trained using Lidstone smoothing, with $\alpha = 1 \times 10^{-6}$, as adopted by Melville et al. [15]. These parameters were chosen since it tends to perform better in text classification tasks in relation to Laplace smoothing [1]. The RF was trained with 64 random decision trees, taking into account the work of Oshiro et al.[18], in which the authors suggest

the number of trees in a RF classifier between 64 and 128. The kind of SVM established for this work is the linear kernel (LSVC), due to its simplicity and shorter training time.

6 RESULTS

Table 1 shows the results for the three trained classifiers. The MNB classifier achieved a great precision and a low recall. The best results were achieved with the LSVC classifier and the RF classifier, with 0.925 and 0.955 of F1-score respectively. The results in our work shows that the SVM classifier algorithm outperformed all the others.

Table 1: The F1 score, Precision and Recall achieved in each classifier

Classifier	MNB		RF		LSVC	
Measures	$\bar{X}$	σ	$\bar{X}$	σ	$\bar{X}$	σ
Precision	**0.986**	0.007	0.883	0.014	0.938	0.027
Recall	0.658	0.040	0.970	0.008	**0.972**	0.005
F1	0.788	0.027	0.925	0.009	**0.955**	0.015

7 CONCLUSION

Hate speech is a growing contemporary problem. In a society, harmonious coexistence and lower rates of violence are desired, so strategies to detect this kind of discourse should be devised. The method in this research can be further developed and used by authorities, backing strategies to address the HS problem.

We collected data from a Brazilian imageboard in order to create a baseline for HS detection in Brazilian Portuguese. We achieved good results in this initial research of HS detection, in two of three classifiers trained for binary classification.

One of the possible limitations of the classifiers is the 'chanspeak' and its relation to semantics. In future work, we intend to create a lexicon-based classifier to detect HS and study of the semantic impact of 'chanspeak' in the HS detection task. We will also attempt to explore the image contents present in the posts, using them as an additional source of analysis.

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